# 16. SETTING UP THE AXIAL PROCESS

DURATION : 16:49 STUDY SHEET

## COURSE SUMMARY

This video teaches the mechanisms of axial process establishment in embryology, focusing on the dynamic between expanding and impanding epiblastic cells. The concepts of the zero point, notochord, and Hensen's node are detailed, illustrating their crucial role in cranio-sacral development. Students will discover how these embryonic structures interact to form a functional unit, essential for understanding all growth and development processes within the framework of biodynamic osteopathy.

## ESTABLISHMENT OF THE AXIAL PROCESS

This movement is driven by a **growth force**. In the **epiblastic** tissue, some cells diverge upwards while others converge.

We observe cells in an **expansion dome** and others in an **impansion dome**. In other words, some cells are in the **convexity** and others in the **concavity**.

The algorithm intervenes: the cells at the front develop much faster than those at the back, with a fulcrum where growth is slow.

Detailing the embryo, the head (cranial, cranio-caudal) is formed from an **impulse**. Below it, another layer of cells forms.

Within this polarity, two phenomena occur in the tissue, visible in the growth rate. The **ectodermal** tissue always grows faster than the tissue below it, capturing information more quickly from the outset.

The **endoblastic** cells, which follow the movement, do not develop in the same way. They do not have the same reference, the same development, nor the same spatio-temporal and energetic information.

There is a point called the **zero point**. In a wave, this point is almost stationary.

In a growth process, it can be represented as follows: at the top is time, and at the bottom is growth. In the centre, the epiblastic tissue is initially almost flat, then quickly takes on an **S** shape.

We observe that the cells along the axis do not grow at the same rate. The upper part grows and begins its expansion, which is the zero point. This represents a rapid multiplication of cells, an accelerated **mitosis**, creating a kind of **tube** below.

This tube is the process of **gastrulation**, called the **notochord**. At this stage, Hensen's node and the tip of the notochord are identical. Hensen's node moves backwards, from the cranial to the caudal region, while the point does not move. This movement leads to the development of the **primitive cranio-sacral axis**.

This system is not yet neural; it is **notochordal**. The tip of the notochord will be the base of the skull, the space for its future development. Hensen's node, at the back, is where the **sacrum** will develop.

It is common to confuse Hensen's node with the closure of the **posterior neuropore**, which is incorrect. In adulthood, the remnant of Hensen's node is found on the sacrum, which includes vertebrae **C1**, **S1**, **S2**, and the descending notochord.

Are the zero point and Hensen's node different? Initially, they are together. Are the future base of the skull and the sacrum together? Yes. They are not really separate, but form a growth wave with a fixed point (the base of the skull) and a mobile system below.

The sacrum is a mobile **fulcrum**, while the base of the skull is a fixed fulcrum. Blais-Schmidt, in classical embryology, speaks of the gastrulation process, but prefers the term **axial process** to explain its establishment.

The zero point is below. The algorithm indicates that the part in the expansion dome will grow much faster than the part in the impansion dome.

Imagine this in a space where the **embryonic pedicle** is present, and trophic information arrives. Inside, a tube forms, thus developing a primitive axial process.

The sacral point is not yet there, but it will arrive. This movement with the zero point and the arriving dome illustrates that the zero point and Hensen's node are the same thing. At the time of this wave, it is the zero point.

When we have the S, it is the zero point. This shows that the sacrum and the occiput are, ultimately, a **functional unit** formed by a movement. The sacrum is directly connected to the base of the skull by this process.

Here we observe a remnant of the notochord, which extends to the base of the skull. This development occurs in the centre of the embryo. At this stage, the anterior part of the embryo is the head, while the small remaining part is the trunk.

The subsequent development of the embryo will change shape, but at this stage, it was in the middle. The vertebrae, the neural tube, and the notochord pass into the vertebrae. Finally, in the embryonic disc, the **nucleus pulposus** will remain.

The axis is present from the outset. To treat disc problems, it is essential to consider what is happening in the **mouth**. The base of the skull, which is the tip of the primitive **autochord**, will be organised by **occlusion**.

The hypophyseal cartilage will become the body of the **sphenoid**, while the parachordal cartilage will become the basilar part of the **occipital**. This set of elements constitutes the **base of the skull**.

It is crucial to see this in three dimensions. At the moment the notochordal process is established, a small formation called the **primitive streak** appears at the level of the embryonic pedicle. This is an energetic field of aspiration that manifests itself.

From this movement, the epiblast will transform into **ectoderm**. Through this aspiration movement, a lateral traction occurs. The great growth of the dome part leads to the appearance of the primitive streak.

This dome forms Hensen's node, which moves backwards by growth. In this phase of development, the lateral cells are drawn towards the centre, forming **bottle cells** that constrict between the tissues.

Between the two, a space opens, creating an aspiration field where cells arrive to form the **primitive mesoderm**. At the moment this third cavity appears, the notochordal process begins, creating the three tissues: **ectoderm**, **mesoderm**, **endoderm**.

The mesoderm begins to form from the primitive streak and the movement of the autochord. Cells are drawn towards the centre to fill the intermediate tissue, forming the **primitive mesoblast**.

It is the epiblastic tissue that will give rise to the mesoderm and the notochord. This process is a powerful wave. The tube created is the notochord, Hensen's node is there, and the primitive streak appears.

As soon as this process begins, the mesoderm forms simultaneously. An important observation is the **nodal flow** which breaks the symmetry of the embryo on a molecular level.

A few years ago, the question arose: why is the heart on the left or right? At the time of this wave, small cilia are observed moving. A liquid called **nodal fluid** becomes a nodal flow.

This flow is closely observed, showing that the cilia perform a circular movement, moving elements from right to left. This movement creates a preferential flow, with small morphogenetic molecules transported by this flow.

These molecules, in the form of small vesicles containing proteins, establish the asymmetry of the embryo. They inform the heart and provide the necessary information for its development.

Thus, at the time of the axial process, a left and a right side, as well as a specific axis, appear. Through this moment, all the mesodermal tissue develops, accompanied by asymmetric morphogenetic activity.

The primitive cranio-sacral axis forms between the **14th** and **21st day**.